Self-scaled bound of the regular 24-cell

We provide an exact certificate for the self-scaled bound of the slack matrix X of the regular 24-cell. We use the matrix `S24cell` in `SlackMat.mat`, with the row and column ordering of the accompanying repository. This is a 24×24 matrix with 144 zero entries, 288 entries equal to 1, and 144 entries equal to 2.

Recall that

$$\mathrm{SSB}(X) = \max_{L \in \mathbb{R}^{24\times 24}} \{\langle L, X\rangle : \langle L, R\rangle \le 1 \text{ for all } R \in \mathcal{R}(X)\},$$

where

$$\mathcal{R}(X) = \{uv^\top : u, v \ge 0,\ uv^\top \le X\}.$$

The script `Certificate24cell.py` checks the enumeration and decomposition below in integer arithmetic.

Proposition 1 (SSB of the regular 24-cell)**.** *The slack matrix X satisfies*

$$\mathrm{SSB}(X) = \frac{32}{3}.$$

An optimal dual certificate is

$$L^*_{ij} = \begin{cases} 0 & \text{if } X_{ij} = 0, \\ 1/36 & \text{if } X_{ij} = 1, \\ 1/108 & \text{if } X_{ij} = 2. \end{cases}$$

Proof. For the lower bound, take L^* as above. Its objective value is

$$\langle L^*, X\rangle = \frac{288}{36} + \frac{2\cdot 144}{108} = \frac{32}{3}.$$

It remains to verify that $\langle L^*, uv^\top\rangle \le 1$ whenever $0 \le uv^\top \le X$.

First, we reduce this verification to a finite enumeration. For a nonzero admissible matrix, normalize its factors so that $\max_i u_i = 1$. Then $0 \le u_i \le 1$ and $0 \le v_j \le 2$. Let $I = \mathrm{supp}(u)$ and $J = \mathrm{supp}(v)$. Since $L^* \ge 0$, we may increase each positive v_j to at least 1, and then each positive u_i to at least $1/2$, without decreasing the objective. These changes preserve $uv^\top \le X$: every entry of X on $I \times J$ is at least 1, while a changed product is at most 1.

On the positive supports, write $u_i = 2^{-a_i}$ and $v_j = 2^{b_j}$, with $a_i, b_j \in [0, 1]$. The constraints corresponding to $X_{ij} = 2$ are automatic, and those corresponding to $X_{ij} = 1$ become $b_j \le a_i$. For $t \in (0, 1)$, define

$$a_i(t) = \mathbf{1}_{\{a_i \ge t\}}, \qquad b_j(t) = \mathbf{1}_{\{b_j \ge t\}}.$$

These binary vectors preserve all inequalities $b_j(t) \le a_i(t)$, and their average over $t \in (0, 1)$ is (a, b). An index with $u_i = 1$ has $a_i = 0$ and remains fixed. The objective $f(a, b) = \sum_{i\in I,\, j\in J} L^*_{ij}\, 2^{b_j - a_i}$ is

convex, so $f(a,b) \le \int_0^1 f(a(t),b(t))\,dt$. Thus, one of these binary points attains at least the original objective. It therefore suffices to consider

$$u_i \in \{0, 1/2, 1\}, \qquad v_j \in \{0, 1, 2\}.$$

Let G be the group of simultaneous row and column permutations constructed by the script. It checks that $|G| = 1152$, that every permutation preserves X, and that the action on the rows is transitive. Since L^* depends only on the value of X_{ij}, these permutations preserve the objective. We may therefore assume $u_1 = 1$. The first row of X contains six zeros, twelve ones, and six twos. Consequently, v_j is zero on the first set, belongs to $\{0,1\}$ on the second, and belongs to $\{0,1,2\}$ on the third. This gives

$$2^{12}3^6 = 2\,985\,984$$

configurations for v. For each configuration, nonnegativity of L^* allows each u_i to be chosen as its largest feasible value:

$$u_i = \begin{cases} 0 & \text{if some } j \text{ has } v_j > 0 \text{ and } X_{ij} = 0, \\ 1/2 & \text{if no such zero occurs and some } j \text{ has } v_j = 2,\ X_{ij} = 1, \\ 1 & \text{otherwise.} \end{cases}$$

Set $C = 108L^*$ and $\widehat{u} = 2u$. Both have integer entries, and

$$216\langle L^*, uv^\top\rangle = \widehat{u}^\top C v.$$

The exhaustive enumeration gives a maximum of exactly 216. Hence L^* is feasible, proving $\mathrm{SSB}(X) \ge 32/3$.

For the upper bound, define two admissible rank-one matrices $R_A = u_A v_A^\top$ and $R_B = u_B v_B^\top$ by

$$\begin{aligned} &(u_A)_1 = 1, \qquad (u_A)_i = 1/2 \quad (2 \le i \le 9), \\ &(v_A)_j = 2 \quad (j \in \{13,14,15,19,20,21\}), \\ &(u_B)_i = 1 \quad (i \in \{1,5,6,19,20,24\}), \\ &(v_B)_j = 1 \quad (j \in \{4,5,12,18,22,23\}), \end{aligned}$$

with all other entries zero. Indices here start at 1. The script checks $0 \le R_A, R_B \le X$ and the exact identity

$$\sum_{g\in G} (3(2R_A)^g + (2R_B)^g) = 864X,$$

where R^g denotes the simultaneous row and column permutation of R. Equivalently,

$$X = \sum_{g\in G} \left(\frac{R_A^g}{144} + \frac{R_B^g}{432} \right).$$

Every term is a nonnegative multiple of an admissible rank-one matrix. For any feasible L, this gives

$$\langle L, X\rangle \le |G| \left(\frac{1}{144} + \frac{1}{432}\right) = 1152 \left(\frac{1}{144} + \frac{1}{432}\right) = \frac{32}{3}.$$

Together with the lower bound, this proves the equality. □

Verification From the repository root, run:

```
python Certificates/Certificate24cell.py
```

The script loads `S24cell` from `SlackMat.mat` and requires NumPy and SciPy. It checks the symmetries, the exhaustive enumeration, and the decomposition in integer arithmetic, without an optimization solver.